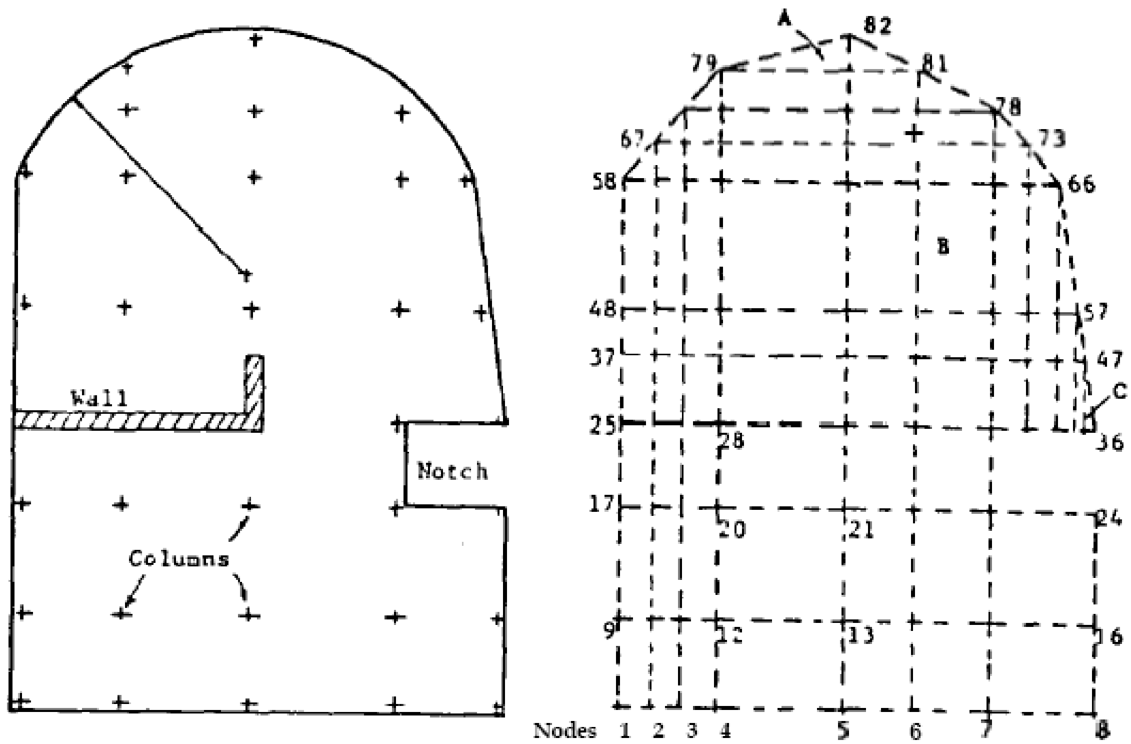
Suggested Analysis and Design Procedures for Combined Footings and Mats

ACI-336-2r16

6.4-Finite element method

This method discretizes the mat into a number of rectangular and/or triangular elements (Fig. 6.5). Earlier programs used displacement functions which purported to produce conforming interelement compatibility (i.e., compatibility both at nodes and along element boundaries between adjacent elements). or For bending, the nodal displacement and slopes in the X- and Y-direction are required. This results in taking partial derivatives of the displacement function. First, however, one must delete one term for the triangle or add two nodes and use the fifteen-term displacement function. Similarly, for the rectangle one must drop three terms or add a node. There are computer programs which drop terms, combine terms, and add nodes. Any of the programs will give about the same computed output so the preferred program is that one most familiar to the user.

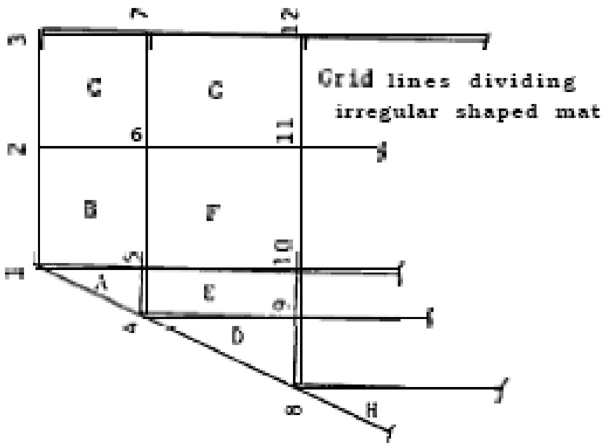


This grid produces 70 elements, 82 nodes, 246 equations
Bandwidth depends on node numbering scheme =3 (largest element node difference + 1)
Here. bandwidth = 3(37 + 25 +1) = 39

a. Mat with wall, notch. and irregular shape including a curved area.

b. Finite element model for mat "a." Curved area has been replaced with a series of straight segments, and triangles are utilized. Gridding shown here is to illustrate method of modeling and may require refinement in some cases.

Fig. 6.2-Mat gridding for either finite grid or finite element method



Compute spring constants K_i as follows:

Let the area of any element such as A, B, C.... = A, B, C....

Let the modulus of subgrade reaction within any element be k_A, k_B, etc.

Then: $K_1 = Ak_A/3 + Bk_B/4$

$K_2 = Bk_B/4 + Ck_C/4$ (typical side spring)

$K_3 = Ck_C/4$ (typical corner spring)

$K_4 = Ak_A/3 + Ek_E/4 + Dk_D/3$

$K_5 = Ak_A/3 + (Bk_B + Ek_E + Fk_F)/4$

$K_6 = (Bk_B + Ck_C + Fk_F + Gk_G)/4$ (typical interior spring)

A computer routine can develop corner, side and interior springs easily but is substantially more difficult for springs with contributions from triangle elements.

A good computer program should be able to compute the typical corner, side and interior springs and allow the user to input by hand springs such as for nodes 1, 4, and 8 above.

Fig. 6.8-Computation of uncoupled Winkler-type soil node springs

6.4.1 *Iso-parametric elements*-An element is of the iso-parametric type if the same function can be used to describe both shape and displacement. Some computer programs suitable for mat (or plate) analysis use this methodology. The method is heavily computation-intensive but has a particular advantage over displacement functions in that any given element may contain a different number of nodes than others as illustrated in Fig. 6.5. Interpolation functions are used together with an approximation for area integration. Good results are claimed with care in using enough interpolation points. Some programs using the iso-parametric formulation may not have the necessary routines to include extra nodes on select elements and in these cases there is usually no advantage of the iso-parametric element over the displacement function approach. Neither of these approaches is exact, unless substantial effort is expended. Cook (1974), Ghali and Neville (1972), and Zienkiewicz (1977) describe formulation of the finite element stiffness matrix.

Advantages of the FEM include:
a. The solution is particularly mathematically efficient.
b. Boundary condition displacements can be modeled.
c. The iso-parametric approach can allow use of midside nodes, which produces some mesh refinement without the large increase in grid lines of the other methods.

Disadvantages of the FEM include:
a. The formulation of the stiffness matrix is computationally intensive.
b. It is very difficult to include a concentrated moment at the nodes as the methodology uses moment per unit of width.
c. Often a statics summation of nodal moments is only approximated. This is often the case where the common elements include triangles or a mix of triangles and rectangles.
d. It requires interpretation of the plate twist moment M_{xv} that is output.
e. It is not directly amenable to increasing the nodal degree-of-freedom from three to six for pile caps as is the finite grid method (FGM).
f. The stiffness matrix is the same size as the FGM but three times the size of the FD method.

The FEM is particularly sensitive to aspect ratios of rectangular elements and intersection angles of triangles.
In practical problems, the designer has little control over these factors other than increasing the grid lines (and number of nodes). For example, Elements A, B, and C of Fig. 6.2 may not give reliable computed results. This deficiency may be overcome for that example by adding grid lines, but this is at the expense of increasing user input and a rapid increase in the size of the stiffness matrix since it increases at three times the number of nodes.

6.5-Column loads

Columns apply axial loads, shears, and moments to a mat. Generally the shear is neglected since its effect would be to slide the foundation laterally. There could also be a tendency to compress the mat between two columns; however, the AE_c/Z term is so large that this movement is negligible and can be ignored. There is no practical way to incorporate horizontal loads into the mat, except at the expense of additional computations. For example, the FGM would require six degree-of-freedom nodes to allow this. This represents prohibitive effort for the results-except for pile caps, where some (or all) of the horizontal shear force may be assumed to be carried by the piles.
The mat may be gridded through the column centers (Fig. 6.6) with any of the methods. If column fixity, zero rotation, at the four column nodes of Fig. 6-6(b) is modeled, the grid lines must be taken through the column faces for the FGM or FEM. If the column lies within the grid spacing, the adjacent nodes are loaded with a portion of the load. The simple beam analogy shown in Fig. 6.6(c) is sufficiently accurate and may be used for both axial load and column moments.

6.6-Symmetry

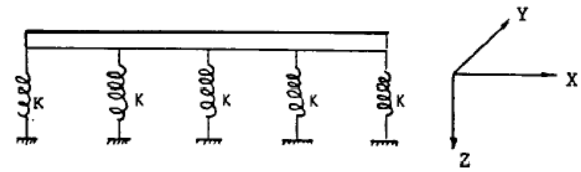
Consideration should be given to any mat symmetry to reduce the total number of nodes to be analyzed. This is critical in the FGM and FEM in particular since I *there* are three equations per node so that a mat with 400 or more nodes will require substantial memory and may require special matrix reduction routines which block the stiffness matrix to and from a disk file.
With symmetry, perhaps only one-quarter or one half of the mat will require modeling. If there is symmetry only for selected load cases and not for others, nothing is gained by attempting to analyze some load cases with a reduced mat plan. Fig. 6.7 illustrates the use of symmetry to reduce a simple plate problem by approximately one-quarter. Success in using symmetry to reduce the problem size depends very heavily on being able to recognize and input the correct nodal boundary conditions along the axes of symmetry. A careful study of whether symmetry applies should be made prior to any modeling, since a substantial amount of engineering and programming effort is often required in developing the element data.
A major factor where computer memory is adequate is whether there is sufficient advantage in utilization of any symmetry to reduce the problem size. Some savings of computer resources are offset with engineering time of identification and input of the extra boundary
conditions. It is very easy to make an error, and careful inspection of the output displacement matrix is necessary.

6.7-Node coupling of soil effects

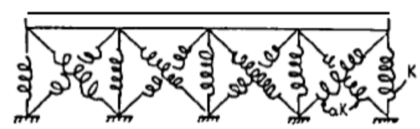
When the mat is interfaced with the ground, the soil effect is concentrated at the grid node for computer analysis. The most common soil-to-node concentration effect produces the so-called Winkler foundation. That is, the modulus of subgrade reaction is used and the concentration is simply

K = Contributory area X k_s (6-7)

This is shown for several element shapes in Fig. 6.8. Since the concentration is a product of area and k_s, the result has the units of a spring and is commonly called a soil spring. Strictly, nodes might be coupled [Fig. 6.9(b)], but this is seldom done, If a Winkler foundation is used, the soil spring K direetly adds (superposition) to every third diagonal term of the stiffness matrix, making very easy to model excessive displacements into the making it soil or mat-soil separation at nodes. This allows reuse of the stiffness matrix by removing the spring as necessary on subsequent cycles. Since the elastic parameters E, and ,4, usually increase with depth from overburden and preconsolidation it appears this may lessen the effect of ignoring coupling (Christian 1976).



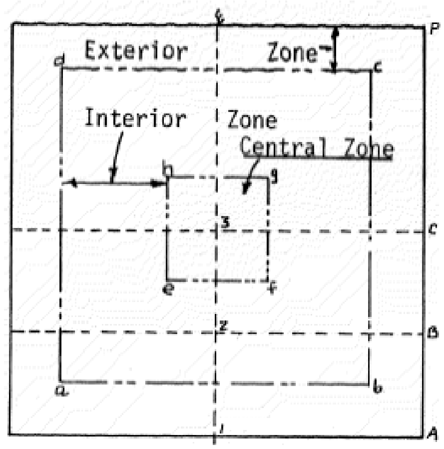
(a) Uncoupled springs



(b) Coupled springs

If coupling were undertaken, the equations to be programmed become much more complicated and, additionally, fractions of the soil concentration appear in off-diagonal terms of the stiffness matrix. If nonlinearity is allowed (soil separation or excessive deformation) the stiffness matrix requires substantial modifications
and it becomes about as easy to completely rebuild it on subsequent cycles.
The most direct and obvious effect of coupling versus noncoupling is that this analysis of a uniformly loaded mat (an oil tank base) will produce a dishing profile across the slab with coupling and a constant settlement profile without coupling. Boussinesq theory
indicates a dishing profile is correct. It is possible to indirectly allow for coupling as follows (Bowles 1984 and Fig. 6.10):

1. Obtain a vertical pressure profile for the mat using any accepted procedure. The Newmark (1935) method in Bowles (1984) might be programmed or the pressure bulbs in Bowles (1982) might be used. This is done at sufficient points so that the mat plan can be subsequently zoned with different values of k,. Use a convenient influence depth of *38* to *5B*, subject to the approval of the geotechnical engineer. Only slightly
different results are obtained using 3*B* versus 10*B* in an elastic system, therefore, a suggested value of 4*B* may be sufficiently accurate.
Table 6.1 provides values at the 5/s points for several mat plans which can he used directly or with linear interpolation for other mat shapes.
2. Numerically integrate the vertical stress profile to obtain the average pressure DO. Designate the edge value as DO,.
3. Compute k_s at any point *i* (refer to Fig. 6.10) as
The furnished value of k_s is taken for the mat perimeter zone.
4. Assign values of k_{si} in the interior as practical for the gridding scheme used for the mat. When doing this take into account how the nodal springs will be computed (refer to Fig. 6.8 and 6.10).

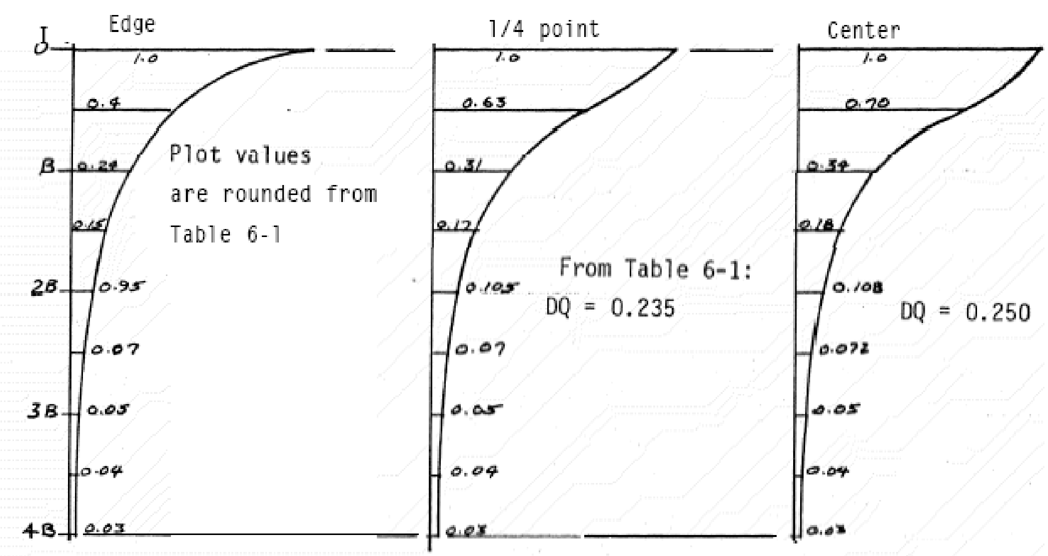


Point 1 has 2 contributing areas of size 1-A- -D-E-1

Point 2 has 4 areas with 2 contributions from 2-B-A-1-2
2-B-D-E-2

Point 3 has 4 contributing areas from 3-C-D-E-3

Possible zones are as shown with outer zone using k_s furnished.
Other zones are bounded by a-b-c-d and e-f-g-h; inner zone is e-f-g-h.



Compute average DQ as:

$$DQ = \frac{0.5B}{4B}\left(\frac{1.0 + 0.028}{2} + 0.40 + 0.24 + 0.146 + 0.095 + 0.066 + 0.048 + 0.036\right)$$

$= 0.193(q_o)$ Using values directly from Table 6-1

For $k_s = 500$: Obtain edge $k_s = 500$; zone 1 = 500(.193/.235) = 410.6 (use 410.)
Central value = 500(.193/.250) = 386.

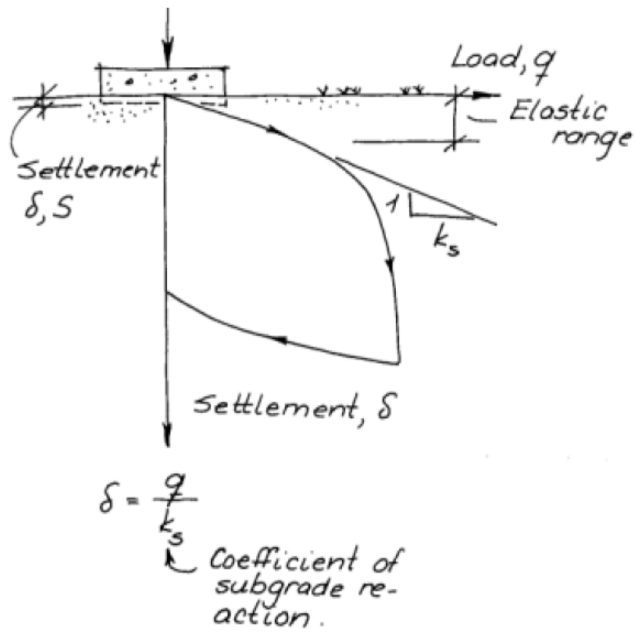
Mat Foundation Analysis and Design

Coefficient of Subgrade Reaction

Nonrigid methods must take into account that both the soil and the foundation have deformation characteristics.

These deformation characteristics can be either linear or non-linear (especially in the case of the soils)

The deformation characteristics of the soil are quantified in the coefficient of subgrade reaction, or subgrade modulus, which is similar to the modulus of elasticity for unidirectional deformation



Definition of Coefficient of Subgrade Reaction

Ks = coefficient of subgrade reaction, units of force/length3 (not the same as unit weight!)

q = bearing pressure settlement

δ = settlement

$$K_s = q/\delta$$

Application of coefficient of subgrade reaction to larger mats

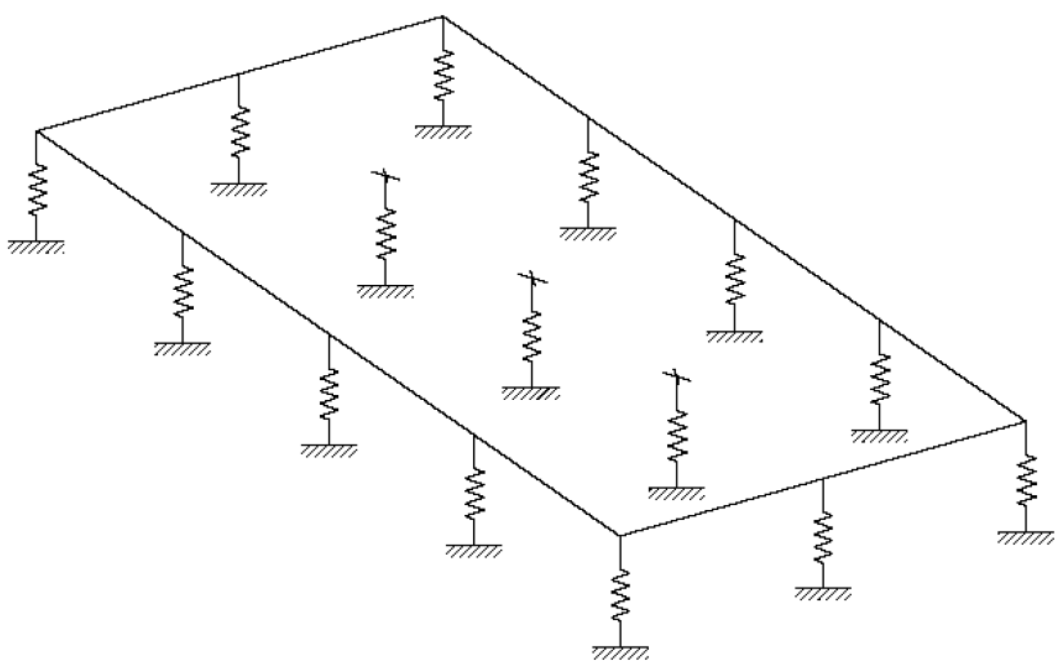
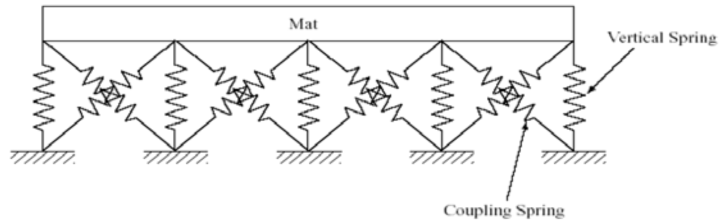


Figure 10.6 The coefficient of subgrade reaction forms the basis for a "bed of springs" analogy to model the soil-structure interaction in mat foundations.

Coupled Method

Ideally the coupled method, which uses additional springs as shown below, is more accurate than the Winkler method
The problem with the coupled method comes in selecting the values of ks for the coupling springs



Pseudo-Coupled Method

An attempt to overcome both the lack of coupling in the Winkler method and the difficulties of the coupling springs
Does so by using springs that act independently (like Winkler springs), but have different ks values depending upon their location on the mat _ Most commercial mat design software uses the Winkler method; thus, pseudo-coupled methods can be used with these packages for more conservative and accurate results

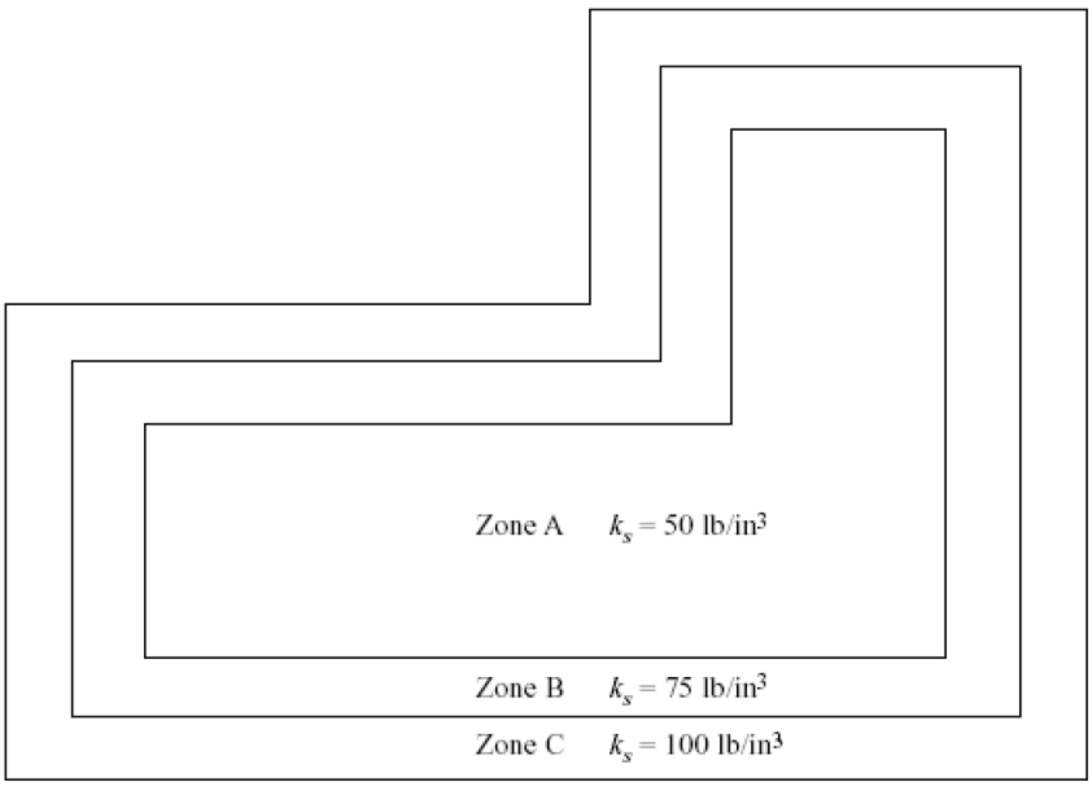


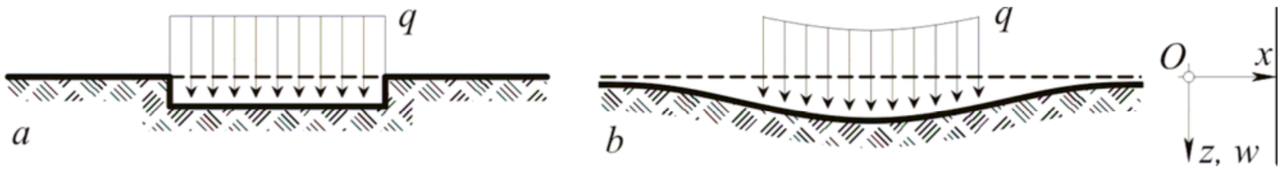
Figure 10.10 A typical mat divided into zones for a pseudo-coupled analysis. The coefficient of subgrade reaction, k_s, progressively increases from the innermost zone to the outermost zone.

(Aa+Ab+Ac).Ko=Aa.Ka+Ab.Kb+Ac.Kc=Ka.(Aa+1.5 Ab+2 Ac)

Divide the mat into two or more concentric zones
The innermost zone should be about half as wide and half as long as the mat.
Assign a ks value to each zone
These should progressively increase from the centre
The outermost zone ks should be about twice as large as the innermost zone
Evaluate the shears, moments and deformations using the Winkler method
Adjust mat thickness and reinforcement to satisfy strength and serviceability requirements

Two parameter Soil (Pasternak foundation model)

Generally, the analysis of bending of beams on an elastic foundation is developed on the assumption that the reaction forces of the foundation are proportional at every point to the deflection of the beam at that point. The vertical deformation characteristics of the foundation are defined by means of identical, independent, closely spaced, discrete and linearly elastic springs. The constant of proportionality of these springs is known as the modulus of subgrade reaction, k_s. This simple and relatively crude mechanical representation of soil foundation was first introduced by Winkler, in 1867 .



Deflections of elastic foundations under uniform pressure: *a* – Winkler foundation; *b* – practical soil foundations.

The Winkler model, which has been originally developed for the analysis of railroad tracks, is very simple but does not accurately represents the characteristics of many practical foundations. One of the most important deficiencies of the Winkler model is that a displacement discontinuity appears between the loaded and the unloaded part of the foundation surface. In reality, the soil surface does not show any discontinuity Historically, the traditional way to overcome the deficiency of Winkler model is by introducing some kind of interaction between the independent springs by visualising various types of interconnections such as flexural elements (beams in one-dimension (1-D), plates in 2-D), shear-only layers and deformed, pretensioned membranes . The foundation model proposed by Filonenko and Borodich in 1940 acquires continuity between the individual spring elements in the Winkler model by connecting them to a thin elastic membrane under a constant tension. In the model proposed by Hetényi in 1950, interaction between the independent spring elements is accomplished by incorporating an elastic plate in threedimensional problems, or an elastic beam in two-dimensional problems, that can deforms only in bending. Another foundation model proposed by Pasternak in 1954 acquires shear interaction between springs by connecting the ends of the springs to a layer consisting of incompressible vertical elements which deform only by transverse shearing . This class of mathematical models have another constant parameter which characterizes the interaction implied between springs and hence are called *twoparameter models* or, more simply, *mechanical models* (Figure 2).

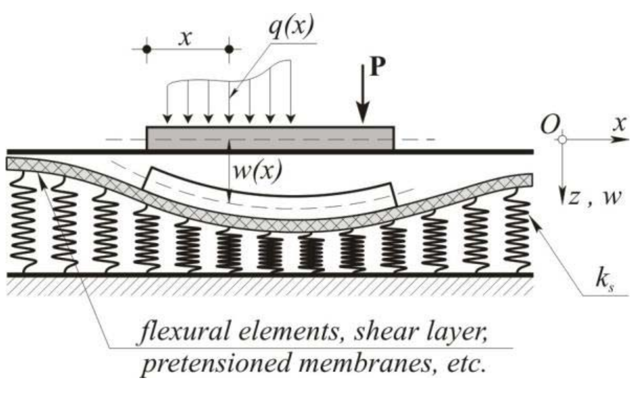


Figure 2. Beam resting on two-parameter elastic foundation.

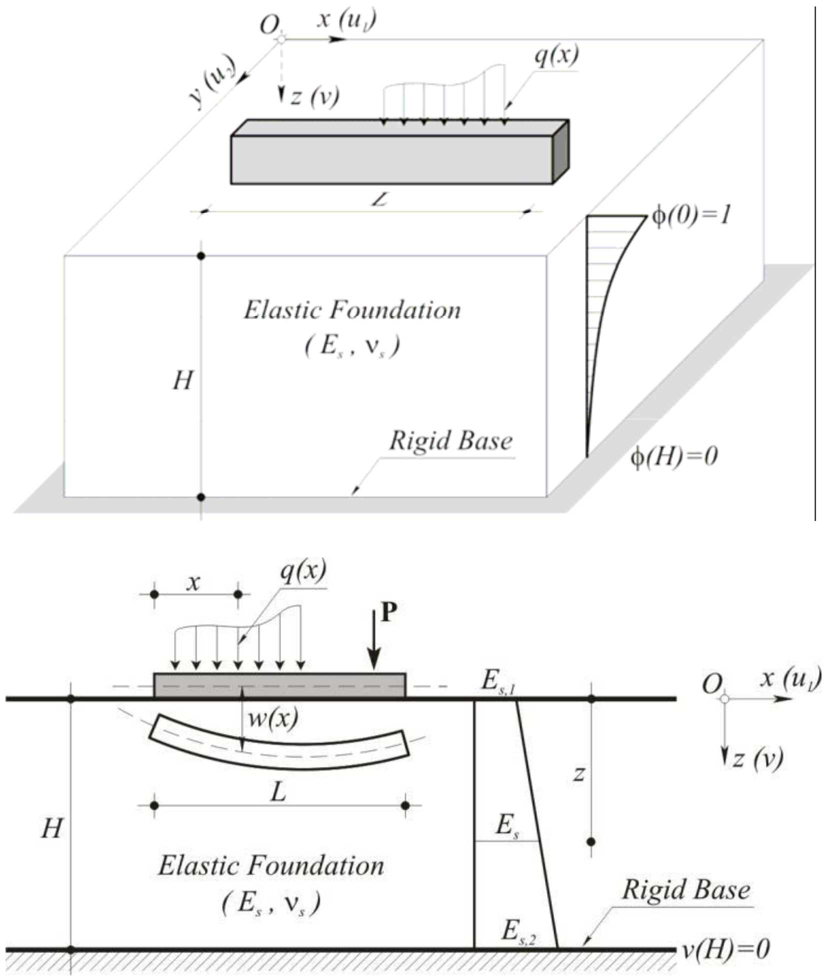


Figure 3. Beam resting on two-parameter foundation.

ESTIMATION OF SUBGRADE SHEAR MODULUS FROM MODULUS OF SUBGRADE REACTION

Hovarth (1989) considered an homogeneous elastic layer of thickness *h*.. which rests on a rigid base and is subjected to a vertical distributed load *p(x, y),* where *x*- and *y-axes* are placed on the free surface. By assuming normal stresses σ_x and σ_y ,shear stress τ_{xy} and horizontal displacements to be zero throughout the layer, Horvarth derived the following relationship between surface deflection *w(x, y)* and applied pressure

$$p(x,\ y) = \frac{E_s}{h_s}\, w(x,\ y) - \frac{E_s h_s}{4}\, \nabla^2 w(x,\ y)$$

where *E.* = elastic modulus of subgrade. This expression is similar to the response equation of the Pasternak foundation Equating the coefficient terms leads to

$$p(x,\ y) = kw(x,\ y) - G_b \nabla^2 w(x,\ y) \qquad G_b = \frac{h_s^2}{4}\, k$$

Eq. gives unrealistic results for large *h*•. Hovarth suggested that an empirical modification be made to the equation by replacing *h.* with a so-called load influence depth *h.* in cases where *h.* is large compared to the size of loaded area, that is

$$G_b = \frac{h_e^2}{4}\, k$$

Unfortunately, there is no simple way of selecting the value of *h.,* which is likely to be a function of, among other factors, the size of loaded area. In the case of rigid highway pavements, the effect of the size of loaded area (i.e., tire contact area) may not be too significant for the following reasons: The load will be distributed through the concrete slab before reaching the foundation; and (2) the tire contact areas of road vehicles do not vary over a wide range as compared to the base of structural footings. In addition, the magnitudes of vehicular wheel loads are small compared to the loads of most structural columns. For most pavement systems under loads, the level of vertical stress decreases rapidly with depth, and falls below 10% of the applied pressure at a depth of 3 times the radius of loaded area (Burmister 1958; Yang 1972; Yoder and Witczak 1975). Beyond this depth, the rate. of change of stress with depth is very small. Therefore, Hovarth's proposal of influence depth in (4) appears to be a reasonable assumption for highway pavements. In the present study, is rewritten as

$$G_b = \theta k$$

where θ = coefficient to be estimated from field measurements, which is the main topic of concern of the present paper.

Values of θ Producing Perfect Pavement Response Predictions for Slabs

Loading position (1)	Range of θ Producing Perfect Prediction (m^2): Deflection prediction (2)	Range of θ Producing Perfect Prediction (m^2): Bending stress prediction (3)
Edge	0.34–1.45[a]	0.35–0.55
Corner	0.35–0.42[b]	0.00–0.46
Interior	0.25–0.35[a]	0.30–1.28

Comparison of Theoretical Solutions with Measured Pavement Responses

<u>Predicted Responses as Function of θ</u>

In the analysis of, and design for, concrete pavement warping, the primary concern is the magnitude of maximum warping stress. Fig. 5 plots the errors of the theoretical solutions of maximum warping stresses for the four test slabs as a function of θ (hence **Gb** *),* based on the model of thin-plate supponed on Pasternak foundation (Shi et al. 1993). The percent absolute errors are computed according to. The results in Fig. 5 clearly show that the predicted magnitudes of maximum warping stress are quite sensitive to the choice of θ value. The results also suggest a decreasing trend of this sensitivity with slabs of larger sizes. The value of θ that give exact maximum warping stress predictions vary from 0.35 to 0.56 m2·

<u>Comparison with Winkler Foundation Solutions</u>

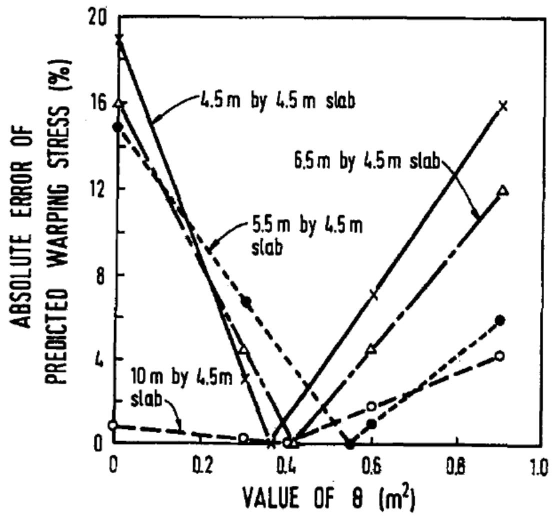
By setting θ = 0, the proposed Pasternak foundation model for warping stress is reduced to one with Winkler foundation. This special case of the model gives a solution that is commonly known as the Bradbury-Westergaard solution for warping stresses (Bradbury 1938; Shi et al. 1993). As can be seen from Fig. 5, the errors of Bradbury-Westergaard solutions (i.e., Winkler foundation solutions corresponding to e = 0) range from 19.0% to about 1.0%. The magnitude of error decreases as the slab size increases. Improved predictions are achieved by incorporating subgrade shear modulus into the foundation model (i.e., having θ > 0 and hence **Gb** > 0). The improvements increase with θ value until the range of θ between 0.35 and 0.55 where the best predictions are obtained. Adopting a θ value of 0.35 m2 as is recommended in provides a comparison of the accuracy of predictions by the Winkler and Pasternak foundation models. The comparison is made in terms of a dimensionless warping stress coefficient, computed by dividing the maximum warping stress by

$(E \cdot \alpha \, .\nabla T)$ where E =elastic modulus of slab; α = thermal coefficient of expansion of concrete; and ΔT = temperature gradient. The advantage of employing Pasternak foundation model is apparent from this comparison, especially for the smaller slab sizes.

Coefficient of Maximum Warping Stress at Center of Slab

Slab size (m) (1)	4.5 × 4.5 (2)	4.5 × 5.5 (3)	4.5 × 6.5 (4)	4.5 × 10 (5)
Measured value	0.479	0.536	0.546	0.584
Winkler solution[a]	0.570	0.622	0.629	0.590
Error (%)	19.0%	16.1%	15.2%	1.0%
Pasternak solution[b]	0.489	0.553	0.579	0.580
Error (%)	2.1%	3.2%	6.1%	0.7%

[a]Solution is obtained from Bradbury-Westergaard solution with Winkler foundation model, which is identical to Pasternak foundation solution with $\theta = 0$.
[b]Solution is obtained from Pasternak foundation solution with $\theta = 0.35\ m^2$.



Errors of Predicted Maximum Warping Stresses in Function of θ

Comparisons of Winkler and Pasternak foundation solutions have been made against field data of full-size slab experiments. Findings summarized in this section are based on Pasternak solutions with θ= 0.35 m2 provides actual measurements for comparisons of load-induced pavement response predictions. Compared with the Pasternak foundation solutions, the Winkler foundation solutions always give higher estimates of maximum deflections and bending stresses. In nearly all the cases analyzed, the Winkler foundation solutions have been found to be conservative, and improved predictions are obtained with Pasternak foundation solutions with a positive value of e. Setting θ = 0.35 m2 , Case Study 1 has shown that the improvements achieved by adopting the Pasternak foundation model range from 1% to 25%.

CONCLUSIONS

Field measurements from two full-size slab experiments have been used to evaluate the application of Pasternak foundation model for engineering analysis of concrete pavement responses under the respective actions of applied loads and thermal loading. The availability of the theoretical solutions for Pasternak foundation models permits direct comparisons of solutions by Pasternak foundation model (with subgrade shear modulus **Gb** > 0) with those of Winkler foundation model (with **Gb** = 0).

The present study shows that Pasternak foundation solutions can provide satisfactory predictions of pavement deflections and bending stresses by having input parameter **Gb** computed as the product of k and a constant coefficient θ.

Using an θ value of 0.35 m2, theoretical solutions using Pasternak foundation model provides better deflection and bending stress predictions with improvements ranging from 1% to 25% as compared with Winkler foundation models.